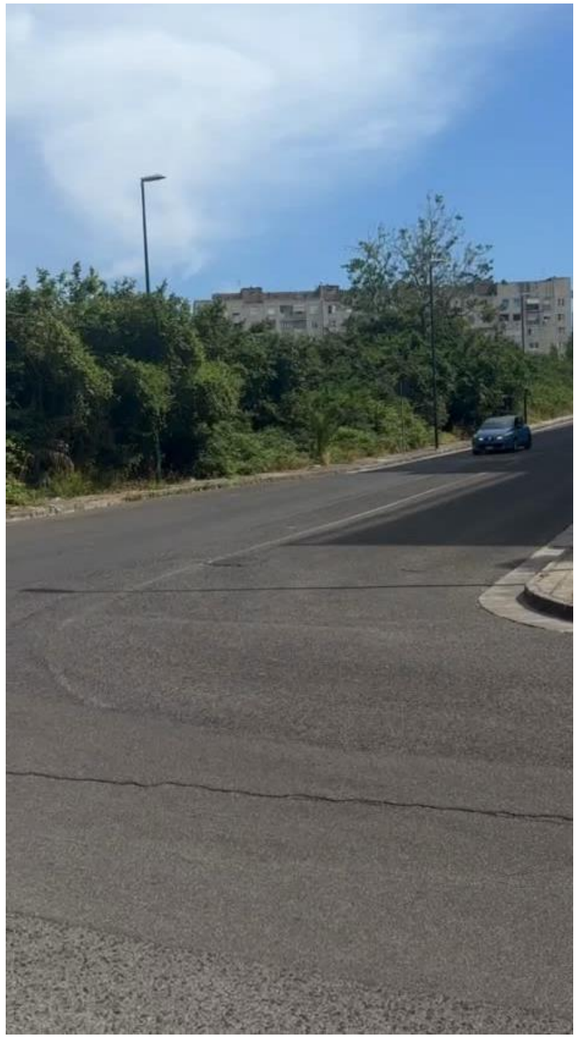
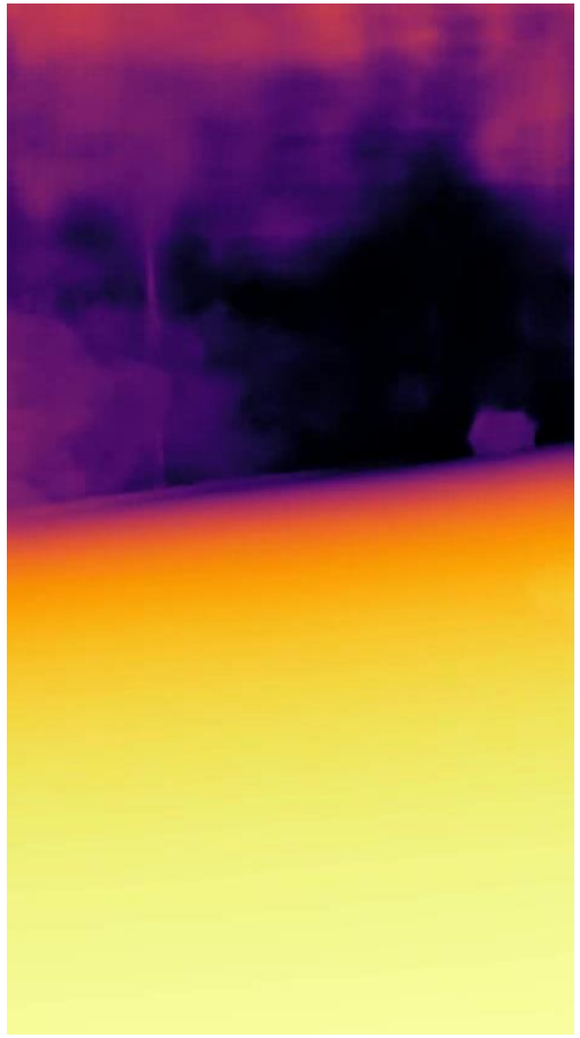
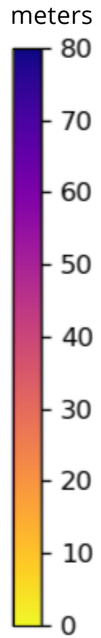
Depth Anything:
Analysis and finetuning in atypical conditions

Romano Davide
Saggiomo Luca
Saccone Matteo

Depth estimation: **what about it?**







Depth estimation: **industrial interest**







TESLA

WAYMO

NVIDIA

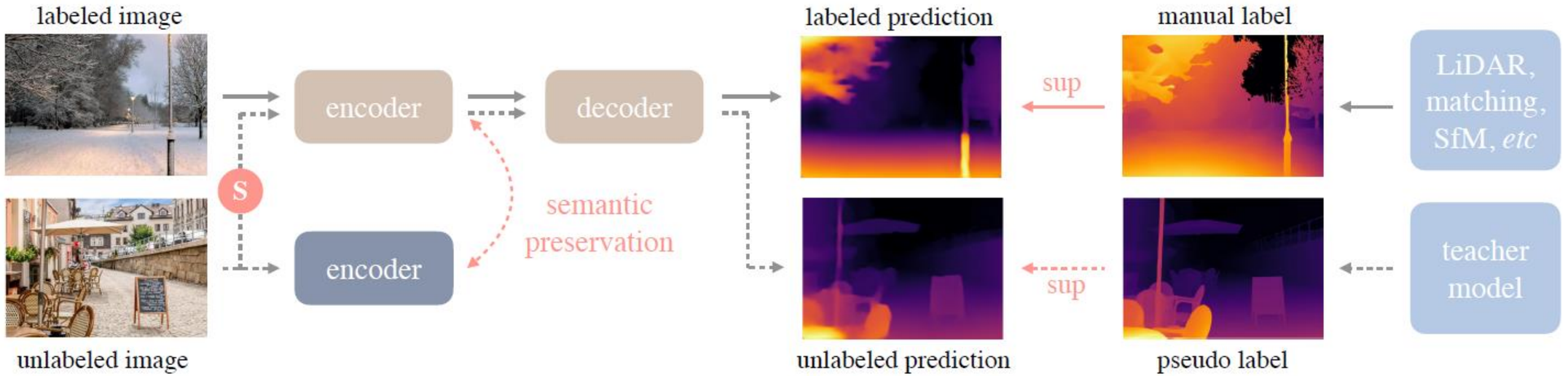
amazon robotics

Boston Dynamics

Meta

Depth Anything: **ins and outs**

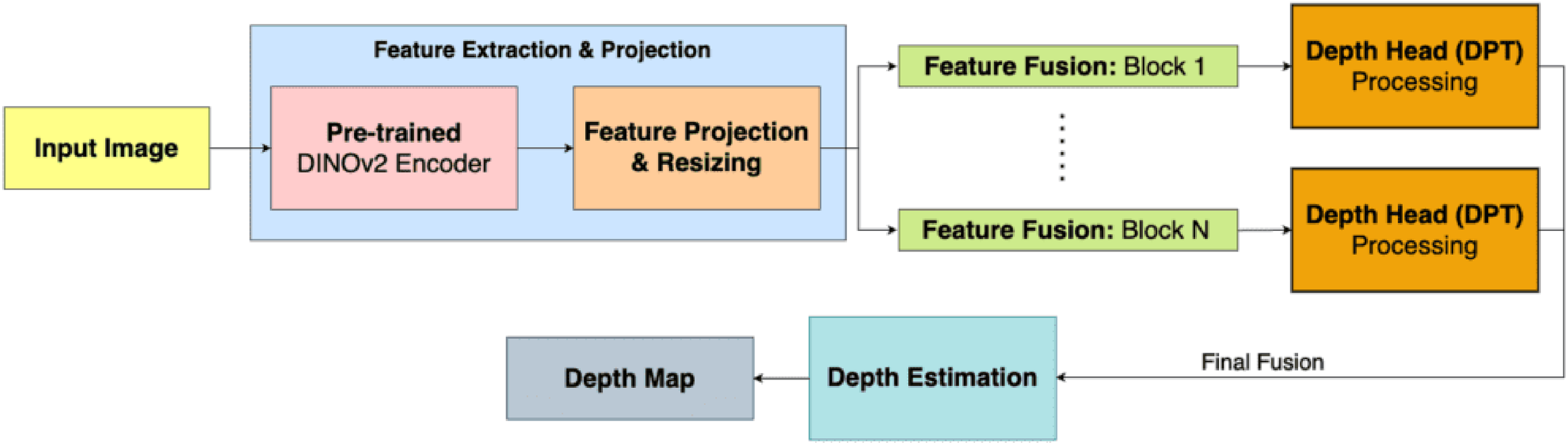
- Teacher-student with *heavy* augmentation approach;
- Encoder-decoder structure;



Yang et. al,"Depth Anything: Unleashing the Power of Large-Scale Unlabeled Data", CVPR 2024.

Depth Anything: **architecture**

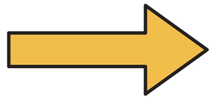
- Vision Transformer
- Dense Prediction Transformer



Depth Anything: just **depth?**

- Segmentation





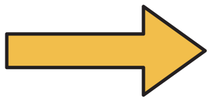


Cheng et. al, "Masked-attention Mask Transformer for Universal Image Segmentation", CVPR 2022.

Depth Anything: just **depth?**

- Segmentation





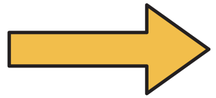


Cheng et. al, "Masked-attention Mask Transformer for Universal Image Segmentation", CVPR 2022.

Depth Anything: just **depth?**

- Segmentation
- Image generation with ControlNet



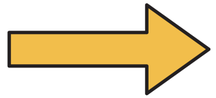


Zhang et. al, “Adding Conditional Control to Text-to-Image Diffusion Models”, 2023 ICCV.

Depth Anything: just **depth?**

- Segmentation
- Image generation with ControlNet





prompt: “spiderman in a city”



Zhang et. al, “Adding Conditional Control to Text-to-Image Diffusion Models”, 2023 ICCV.

Metric depth estimation

- In *safety critical* contexts precise metric depth estimation is imperative for spatial awareness and **operational reliability**.
- Our work uses *Depth Anything V2 Metric Outdoor*, the second version of Depth Anything finetuned for metric depth estimation.

Yang et. al, "Depth Anything V2", NeurIPS 2024.

Depth *Anything?*

Depth ~~Anything~~

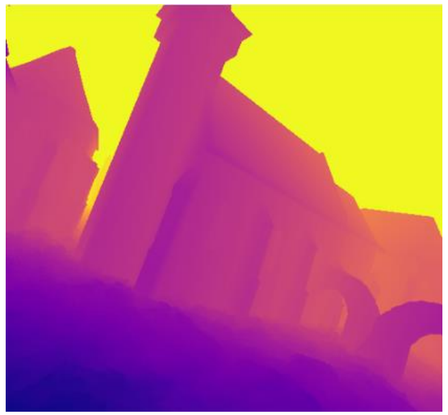
- The model fails in some scenarios

original image



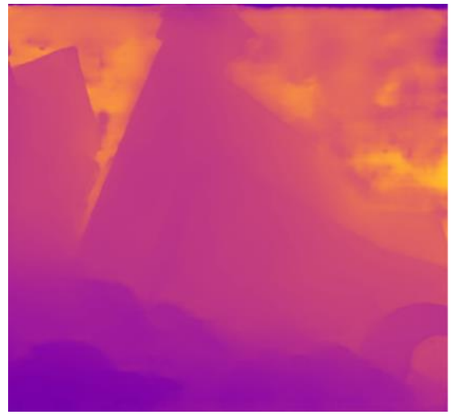


ground truth

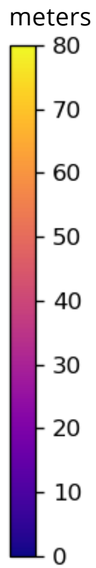




prediction







Our goals

- Finetune the model to improve predictions in *atypical conditions*;

- Emphasis was placed on images with:
  - Weather effects: raining, snowing and fog;
  - Debris;
  - Poor lightning;
  - Aggressive camera movements.

TartanAir **dataset**

- *Photorealistic synthetic* data;
- Multimodal ground truth;
- Focus on *edge cases;*

Developed for the *CVPR 2020 TartanAir SLAM Challenge* to test Visual SLAM robustness under extreme conditions where standard benchmarks, like KITTI, fail.

Wang et. al, "TartanAir: A Dataset to Push the Limits of Visual SLAM", IROS 2020.

TartanAir **scenes**

- The Dataset is composed by various scenes, each with different characteristics

neighborhood



gascola



abandonedfactory_night



westerndesert



oldtown



seasonforest_winter



endofworld



TartanAir **split**

- *Train scenes:*
  - Neighborhood;
  - Season forest winter;
  - Abandoned factory night;
  - Western desert.

- *Validation scenes:*
  - Gascola;
  - Oldtown.

- *Test scenes:*
  - End of world.

TartanAir **problems**

- High temporal correlation since the images are frames of continuous video sequences.
- Poor diversity in images belonging to the same scene.

TartanAir **problems**

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  - **Fixed stride temporal subsampling.**
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TartanAir **problems**

- High temporal correlation since the images are frames of continuous video sequences.
  - **Fixed stride temporal subsampling.**
- Poor diversity in images belonging to the same scene.
  - **Aggressive data augmentation.**

Results on test

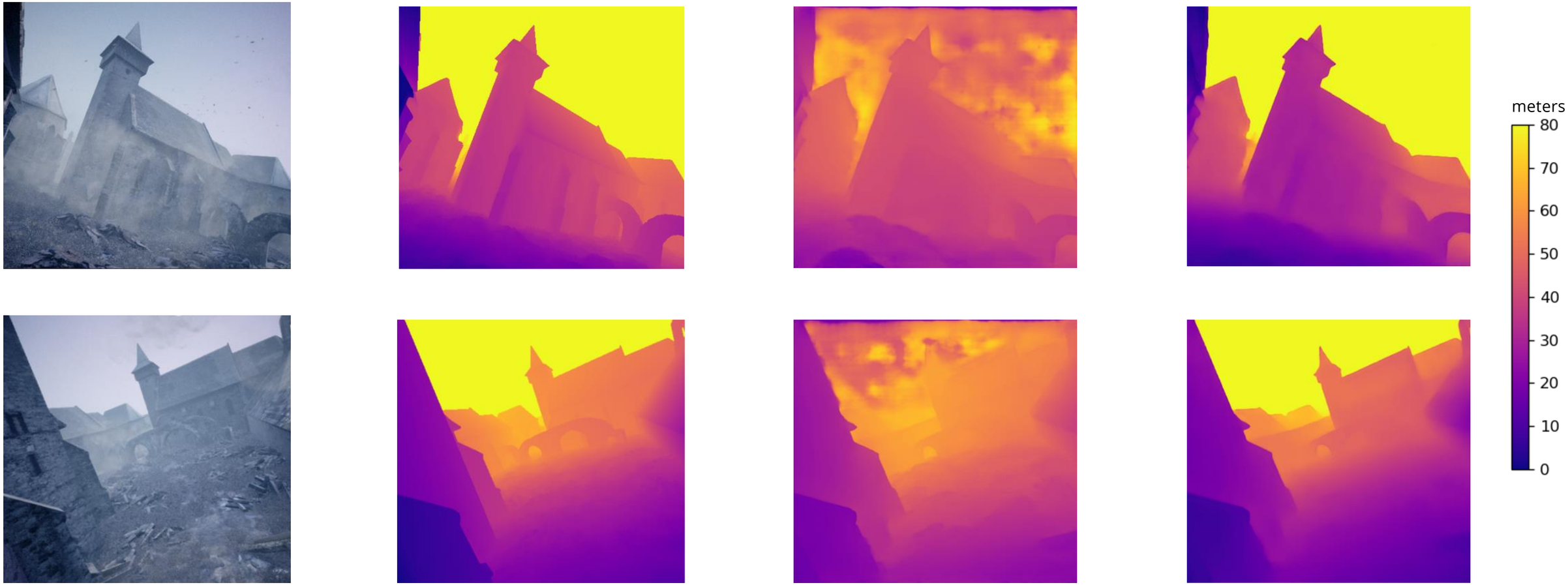
- We evaluated the performance using three metrics:
  - AbsRel, RMSE and δ_1.

Metric	Base values	Finetuned values
AbsRel	0.95	**0.15**
RMSE	21.64	**2.79**
δ_1	0.12	**0.71**

Test on TartanAir images

- The finetuned prediction is *cleaner.*

original image | ground truth | prediction - base | prediction - finetuned



Test on non-synthetic images

- How does our model perform in a real-world scenario?

original image



prediction - base



Test on non-synthetic images

- Encouraging qualitative results.

original image



prediction - finetuned



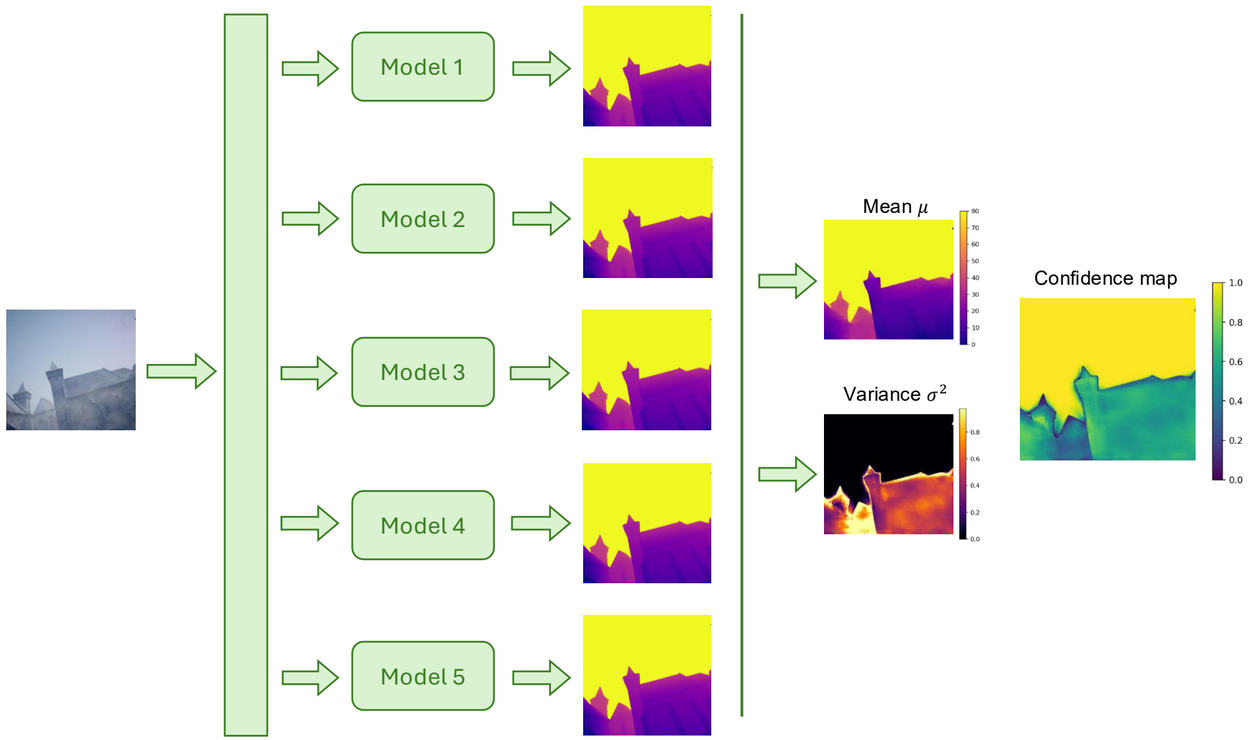
Is the model *trustworthy?*

Model confidence

- AI models have limitations and can fail;
- In critical context this is **unacceptable**;
- We need to evaluate the *confidence interval* of the produced depth values.

Our approach: **Deep Ensemble**

- Combining M independent models with **small variations** in training to regularize predictions and extract spatial model uncertainty.



Lakshminarayanan et. al, "Simple and Scalable Predictive Uncertainty Estimation using Deep Ensembles", NIPS 2017

How did we diversify the models?

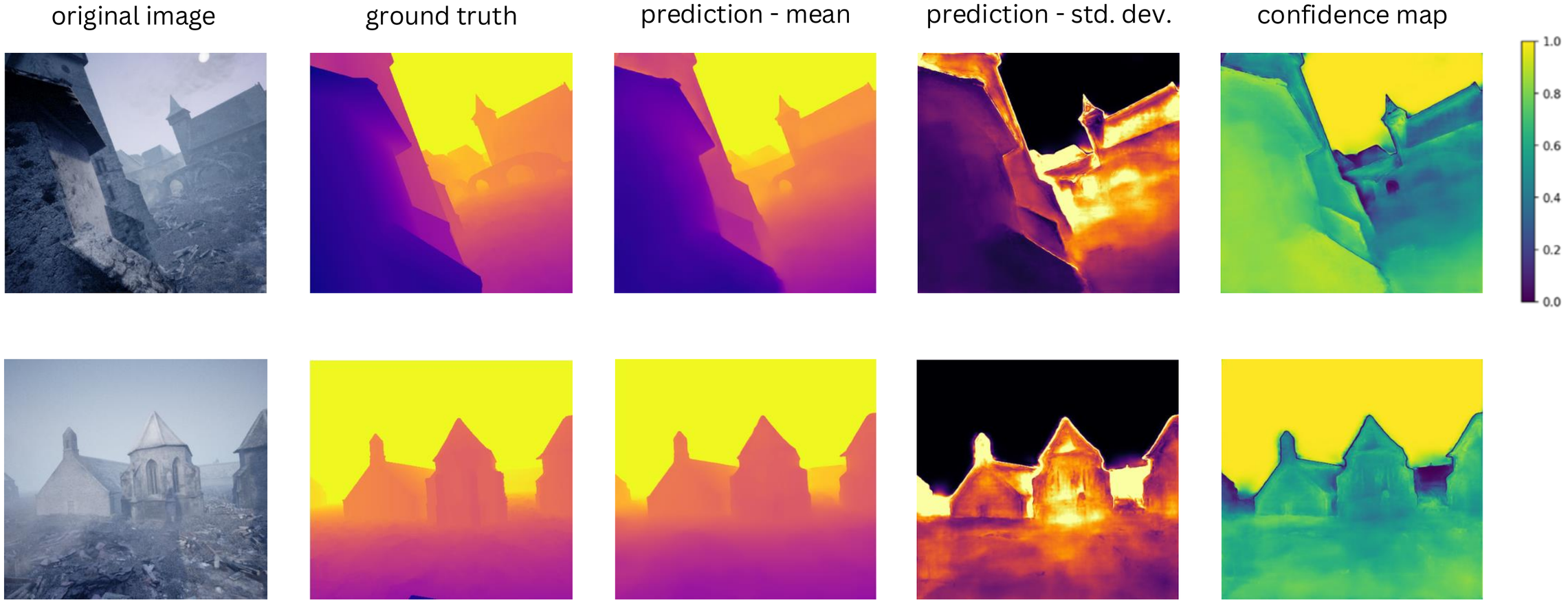
- Learning rate differentation:
  - Different schedulers (fixed learning rate and cosine annhealing);
  - Varying warmup periods.

- Different Gradient-loss weights.

- Different random seeds.

Results

- Models show more uncertainty on edges



Future developments

- Train with non-synthetic datasets like SubT-MRS (large robotic dataset born with DARPA SubT challenge)

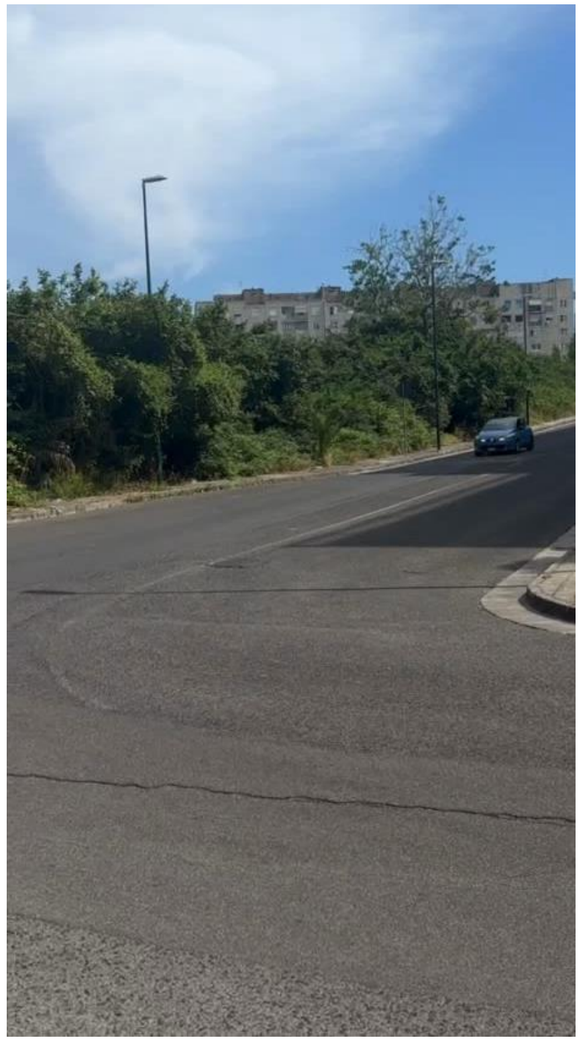
- Edge AI optimization

- Improve confidence map reliability:
  - Transitioning to probabilistic training via Negative Log-Likelihood (NLL) losses, following Lakshminarayanan et. al

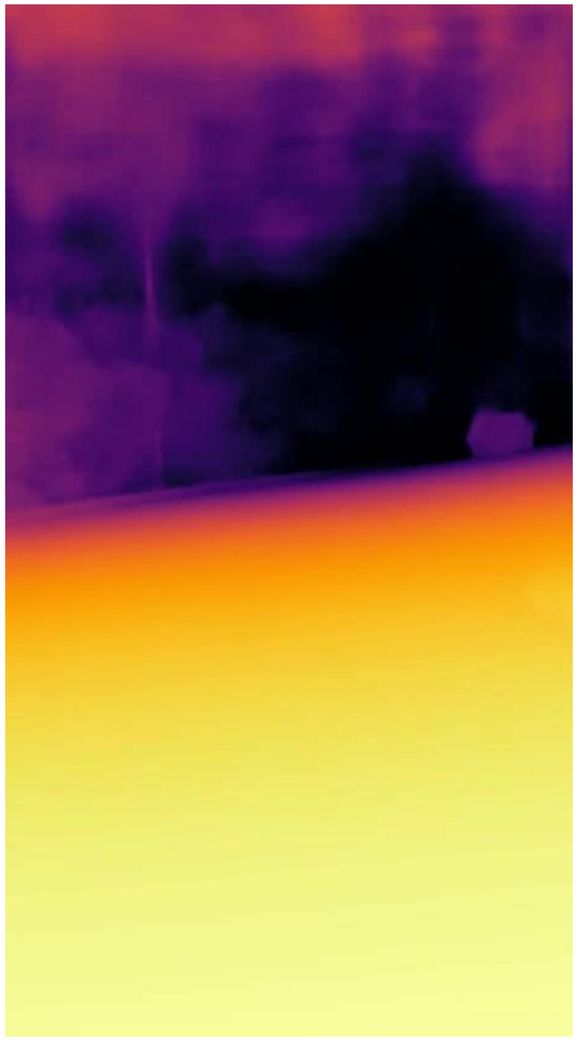
One more thing...

Our video

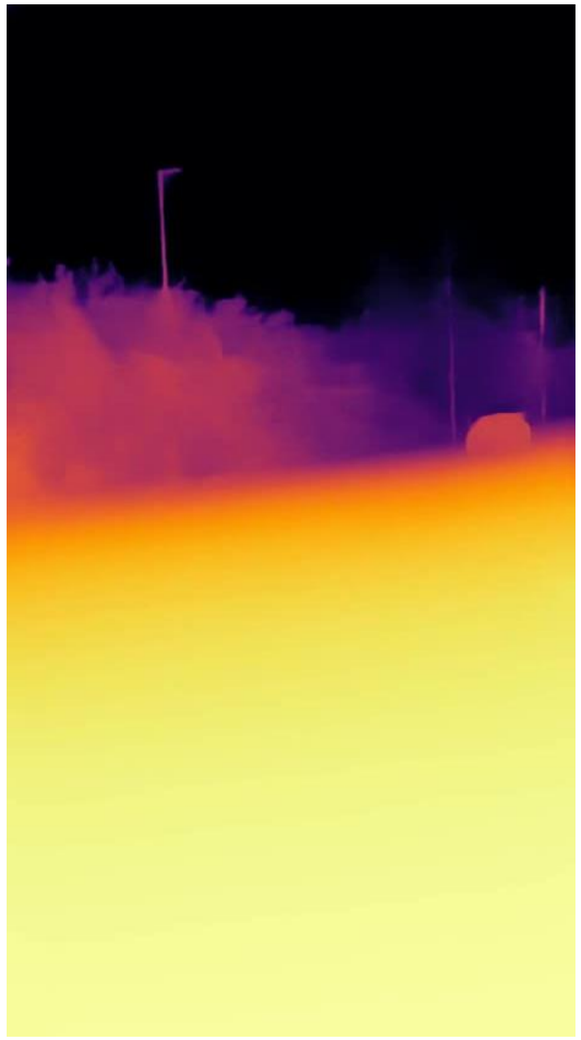
original video

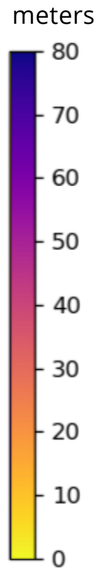


prediction - base



prediction - finetuned





Deep thanks!



